

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 – Present
	Yale University , New Haven, CT Graduate Research Assistant	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018

EDUCATION	Yale University , New Haven, Connecticut, USA	
	▪ Ph.D. in Astronomy with Distinction	Aug 2020 – May 2025
	• Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures	
	• Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	
	▪ M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	▪ B.S. in Astrophysics	Aug 2016 – Jun 2020
	▪ Minor in Computer Science	

PUBLICATIONS 177 citations. h -index = 8.
Italics: Student directly supervised by T. Lu

FIRST-AUTHOR

- [8] **Lu, T.**, Millholland, S.C., Rice, M., Black, B., Bardalez Gagliuffi, D., Balmer, W., Pueyo, L., Giovinazzi, M., and Brandt, T. in review.
- [7] **Lu, T.**, Hernandez, D.M., Rice, M. and Su, Y. in review.
- [6] **Lu, T.**, Tajer, H., Hernandez, D.M., Rein, H., *Liu, Y.* and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* **9**, 5
- [5] **Lu, T.**, Li, G., Cassese, B. and Lin, D.N.C., 2025. [The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet](#). *ApJ* **980**, 39
- [4] **Lu, T.**, An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T., 2025. [Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11](#). *ApJ* **979**, 218
- [3] **Lu, T.**, Hernandez, D.M. and Rein, H., 2024. [TRACE: a Time-Reversible Algorithm for Close Encounters](#). *MNRAS* **533**, 3708
- [2] **Lu, T.**, Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. [Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework](#). *ApJ* **948**, 41
- [1] **Lu, T.** and Laughlin, G., 2022. [Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#). *PSJ* **3**, 221

STUDENT-LEAD

- [4] *Yang, E.*, **Lu, T.**, Yahalomi, D., and Winn, J. in review.
- [3] *Im, H.*, **Lu, T.**, Rice, M., Tran, Q., Li, G. and Naoz, S., in press. [Observational and Dynamical Constraints on an Unseen Outer Perturber in the GJ 436 Hot Neptune System](#).
- [2] *Liu, Y.*, **Lu, T.** and Rice, M., 2025. [Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94](#). *ApJL* **997**, L41
- [1] *Liu, Y.*, **Lu, T.** and Rice, M., 2025. [The Formation of Double Hot Jupiter Systems Through ZLK Migration](#). *ApJ* **986**, 103

SIGNIFICANT CONTRIBUTION

- [5] Yahalomi, D.A., **Lu, T.**, Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A., and Rice, M., 2026. [The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry.](#) *ApJL* **999**, L9
- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides.](#) *PSJ* **6**, 252
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System.](#) *AJ* **169**, 22
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX.](#) *JOSS* **9**, 100
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters](#) *ApJ* **921**, 105

OTHER CO-AUTHOR

- [3] Tajer, H., Wang, J., Childs, C., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury's Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations.](#) *ApJ* **998**, 46
- [2] Bardalez Gagliuffi, D.†, Balmer, W.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System.](#) *ApJL* **988**, L18
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e.](#) *RAA* **25**, 015003

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudon, E., **Lu, T.**, Hsieh, C., O'Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers.](#) *Bulletin of the AAS* **56**, 2024i004

†: Joint first author

AWARDS & FELLOWSHIPS

▪ DDA/AAS Raynor L. Duncombe Student Research Prize	2025
▪ Flatiron Research Fellowship	2025
▪ Flipped Science Fair Best Poster Explanation	2024
▪ AAS International Travel Grant	2024
▪ Graduate Student Assembly Conference Travel Fellowship	2024
▪ National Science Foundation GRFP (<i>Honorable Mention</i>)	2022
▪ Joan and Arnold Seidel Griffith Observer Science Writing Contest (<i>Second Place</i>)	2022
▪ Alan Porter Memorial Foundation Fellow	2019
▪ Summer Undergraduate Research Fellow	2018

ORAL PRESENTATIONS & TALKS

▪ Monday Afternoon Talk, MIT (<i>Cambridge, MA</i>)	Apr 2026
▪ Astronomy & Planetary Science Colloquium*, NAU (<i>Flagstaff, AZ</i>)	Feb 2026
▪ Astronomy Seminar*, AMNH (<i>New York City, NY</i>)	Feb 2026
▪ International Conference on Exoplanets and Planet Formation (<i>Shanghai, China</i>)	Dec 2025
▪ Powell Group Meeting, University of Chicago (<i>Chicago, IL</i>)	Oct 2025
▪ 246th AAS Meeting Dissertation Talk (<i>Anchorage, AK</i>)	Jun 2025
▪ Boston Area Planetary Science Meeting, MIT (<i>Cambridge, MA</i>)	May 2025
▪ New York Area Exoplanets Meeting II, CCA (<i>New York City, NY</i>)	May 2025
▪ 56th DDA Meeting, Georgia Institute of Technology (<i>Atlanta, GA</i>)	May 2025
▪ ML in Planetary Dynamics Workshop*, CCA (<i>New York City, NY</i>)	Apr 2025
▪ Tea Talk, Indiana University (<i>Bloomington, IN</i>)	Feb 2025
▪ ITC Luncheon, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Pizza Lunch, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>)	Nov 2024
▪ Planetary Lunch Seminar, Massachusetts Institute of Technology (<i>Cambridge, MA</i>)	Nov 2024
▪ Astronomy Seminar, Columbia University (<i>New York City, NY</i>)	Oct 2024
▪ Astronomy Seminar, Stony Brook University (<i>Stony Brook, NY</i>)	Oct 2024
▪ CRA Seminar, Georgia Institute of Technology (<i>Atlanta, GA</i>)	Oct 2024

	Planetary Science Seminar, California Institute of Technology (<i>Pasadena, CA</i>)	Oct 2024
	Lunch Seminar, UCLA (<i>Los Angeles, CA</i>)	Sep 2024
	Exoplanet Group Meeting, UCSB (<i>Santa Barbara, CA</i>)	Sep 2024
	FLASH Talk, UCSC (<i>Santa Cruz, CA</i>)	Sep 2024
	Star and Planet Lunch, University of Hawai'i (<i>Honolulu, HI</i>)	Aug 2024
	CT Exoplanet Picnic, Wesleyan College (<i>Wesleyan, CT</i>)	Jul 2024
	Emerging Researchers in Exoplanet Science IX, Cornell University (<i>Ithaca, NY</i>)	Jul 2024
	REBOUND 24 Meeting* (<i>Virtual</i>)	Jul 2024
	<i>Arbitrary ODE Workshop</i>	
	REBOUND 24 Meeting* (<i>Virtual</i>)	Jul 2024
	<i>tides_spin</i>	
	New York Area Exoplanets Meeting I, CCA (<i>New York City, NY</i>)	May 2024
	55th DDA Meeting, University of Toronto (<i>Toronto, Canada</i>)	May 2024
	Makino Lab Seminar, Kobe University (<i>Kobe, Japan</i>)	Jul 2023
	Exoplanet Seminar, Nanjing University (<i>Nanjing, China</i>)	Jul 2023
	54th DDA Meeting, Michigan State University (<i>East Lansing, MI</i>)	May 2023
	Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>)	Mar 2023
	Astronomy Seminar, Monash University (<i>Monash, Australia</i>)	Feb 2023
	240th AAS Meeting (<i>Pasadena, CA</i>)	Jun 2022
	*Invited Talk	
POSTER PRESENTATIONS	Emerging Researchers in Exoplanet Science, Princeton University (<i>Princeton, NJ</i>)	Jun 2025
	Exoplanets V Meeting (<i>Leiden, The Netherlands</i>)	Jun 2024
	Extreme Solar Systems (<i>Christchurch, New Zealand</i>)	Jun 2024
	2022 AGU Meeting (<i>Chicago, IL</i>)	Dec 2022
	Exoplanets IV Meeting (<i>Las Vegas, NV</i>)	May 2022
	2019 AGU Meeting (<i>San Francisco, CA</i>)	Dec 2019
	233rd AAS Meeting (<i>Seattle, Washington</i>)	Jan 2019
TEACHING	ASTR 210: Stars and Their Evolution	New Haven, CT, USA
	Yale University Teaching Fellow	Spring 2023
	ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
	Yale University Teaching Fellow	Fall 2022
	ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
TEACHING	Yale University Teaching Fellow	Spring 2021
	ASTR 155: Introduction to Astronomical Observing	New Haven, CT, USA
	Yale University Teaching Fellow	Fall 2021
	Dean's Tutor	Pasadena, CA, USA
	Tutoring Caltech students in Physics, Astronomy and Computer Science	2017-2020
OUTREACH	Warrior-Scholar Program	New Haven, CT, USA
	Research Project Leader	Summer 2023
	<i>One week intensive boot camp preparing military veterans for undergraduate degrees</i>	
	Yale OpenLabs	
	Local outreach organization dedicated to educating middle school children on STEM topics. Receptient of the Seton Elm-Ivy Award.	
	Public Relations Officer	2023
	Regular Outreach Speaker	2022 – 2025
	Science Cafe 2024 Speaker: <i>What if 'Cotton Candy' Planet are Actually Ring Pops?</i>	
	Flipped Science Fair Best Poster Explanation: <i>Probing the Shapes of Planets</i>	
	Science Cafe 2022 Speaker: How We Discover Exoplanets	
	Astronomy on Tap	
	New Haven branch of outreach program aimed at engaging local community with astronomy research	
	Social Media Director	2022

Science in the News

Local outreach organization dedicated to educating the greater community on STEM topics.

- Outreach Speaker 2021
 - [Hidden Things: Pulling Back the Veil on the Universe](#)

RESEARCH MENTORING

Yurou (Nina) Liu* Jun 2023 –
Alexander P. Hixon Summer Undergraduate Research Fellow Yale University
Project: *The Formation of Double Hot Jupiter Systems Through ZLK Migration*

Haedam Im May 2025 –
Dorrit Hoffleit Undergraduate Research Fellow Massachusetts Institute of Technology
Project: *Hunting for a Hidden Planet in the GJ 436 System*

Eritas Yang Oct 2025 –
Graduate Student Princeton University

Vighnesh Nagpal Aug 2025 –
Graduate Student University of Chicago

Haniyeh Tajer May 2025 –
Graduate Student Ohio State University

Jeremiah Reynoso Jun 2023 – Oct 2023
Summer Undergraduate Research Fellow Morehouse College

*Primary Research Advisor

INFORMAL MENTORING

Chris Santiago 2022 – 2024
Astro Sibbs Yale University

Kylyn Smith 2023 – 2024
Astro Sibbs Yale University

Sally Jiang (current Ph.D. Candidate, Columbia University) 2022 – 2023
Astro Sibbs Yale University

TELESCOPE PROPOSALS (CO-I)

James Webb Space Telescope – 44.7 hours Cycle 5 GO
ID: 11253. Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff. *PI: Ben Cassese*

Keck Observatory (KPF) – 0.5 nights Yale 2026A
Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints. *PI: Quang Tran*

Keck Observatory (KPF) – 2 nights Yale 2024B
The Search for Double Hot Jupiter Systems. *PI: Malena Rice*

Keck Observatory (HIRES) – 2 nights Yale 2023A
Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2). *PI: Emma Louden*

Keck Observatory (HIRES) – 2 nights Yale 2022B
Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1). *PI: Emma Louden*

PROFESSIONAL SERVICE

Science Organizing Committee – 57th DDA Meeting June 2026
Division of Dynamical Astronomy, University of Chicago

Science Organizing Committee Chair – DSPPS Jan 2026
A Dynamical Synthesis of Planetary Systems, New Haven

Organizing Committee – ERES X July 2025
Emerging Researchers in Exoplanet Science Symposium, Princeton University

Science Organizing Committee – ERES VIII June 2023
Emerging Researchers in Exoplanet Science Symposium, Yale University

Reviewer
NASA XRP, NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS 2024 – Present

Main Co-Organizer – Yale Exoplanets and Stars Seminar 2023 – 2025
Weekly Seminar Series

	Committee Member – ACDC <i>Yale University Astronomy Climate and Diversity Committee</i>	2022 – 2025
MEDIA	Yale Daily News	Jun 2025
	New study offers a double dose of ‘hot Jupiters’	
	Popular Science	Oct 2024
	What if ‘cotton candy’ planets are actually Ring Pops?	
	AAS Author Journal Series	Nov 2022
	Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9	
COMMUNITY ENGAGEMENT	New Haven Road Runners – New Haven, CT, USA	
	<i>Local USATF sanctioned running club</i>	
	▪ Public Relations Officer	Jul 2023 – Jul 2024
	▪ Racing Team Director / Elite Athlete Coordinator	Jul 2022 – Jun 2025
	NCAA Athlete – Pasadena, CA, USA	2016 – 2020
	<i>NCAA Division III Cross Country/Track & Field at Caltech</i>	
	Avery House – Pasadena, CA, USA	
	<i>Student Government</i>	
	▪ Athletics Manager	2017 – 2018
	▪ Underclassmen Counselor	2017 – 2020

[CV compiled on 2026-04-17]